Code Quality

git

1. GIT BASICS
2. INTRO TO GIT
3. GIT SETUP
4. BRANCHING STRATEGY
5. APPLYING AI ACROSS THE GIT WORKFLOW
6. WORKSHOP

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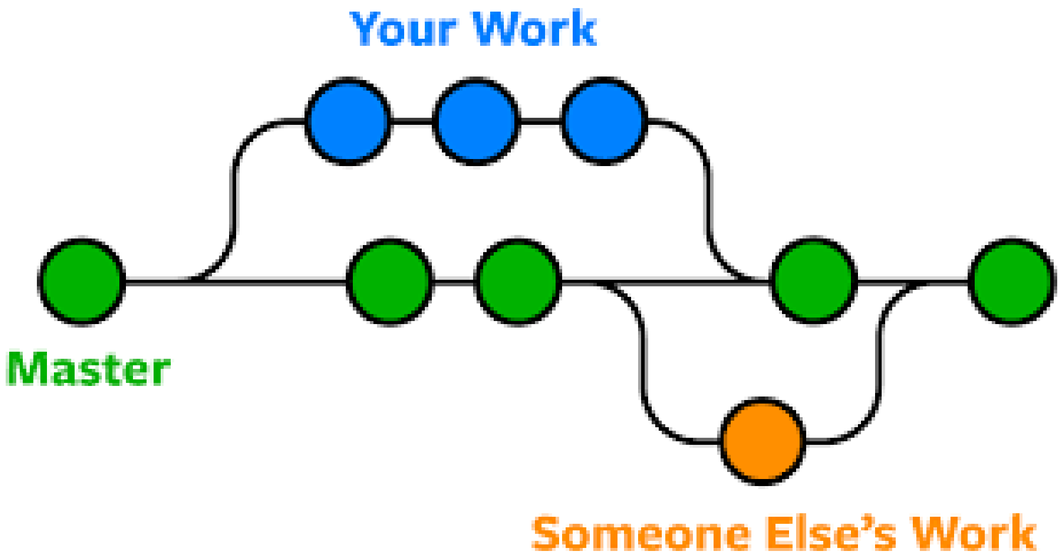
GIT BASICS

WHAT IS GIT?

- Distributed Version Control System (DVCS)
- Tracks changes in files and source code
- Enables collaboration across teams

Why Version Control Matters:

- Keeps history of changes
- Enables rollback to previous versions
- Prevents code conflicts
- Supports parallel development



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INTRO TO GIT

What is a Repository?

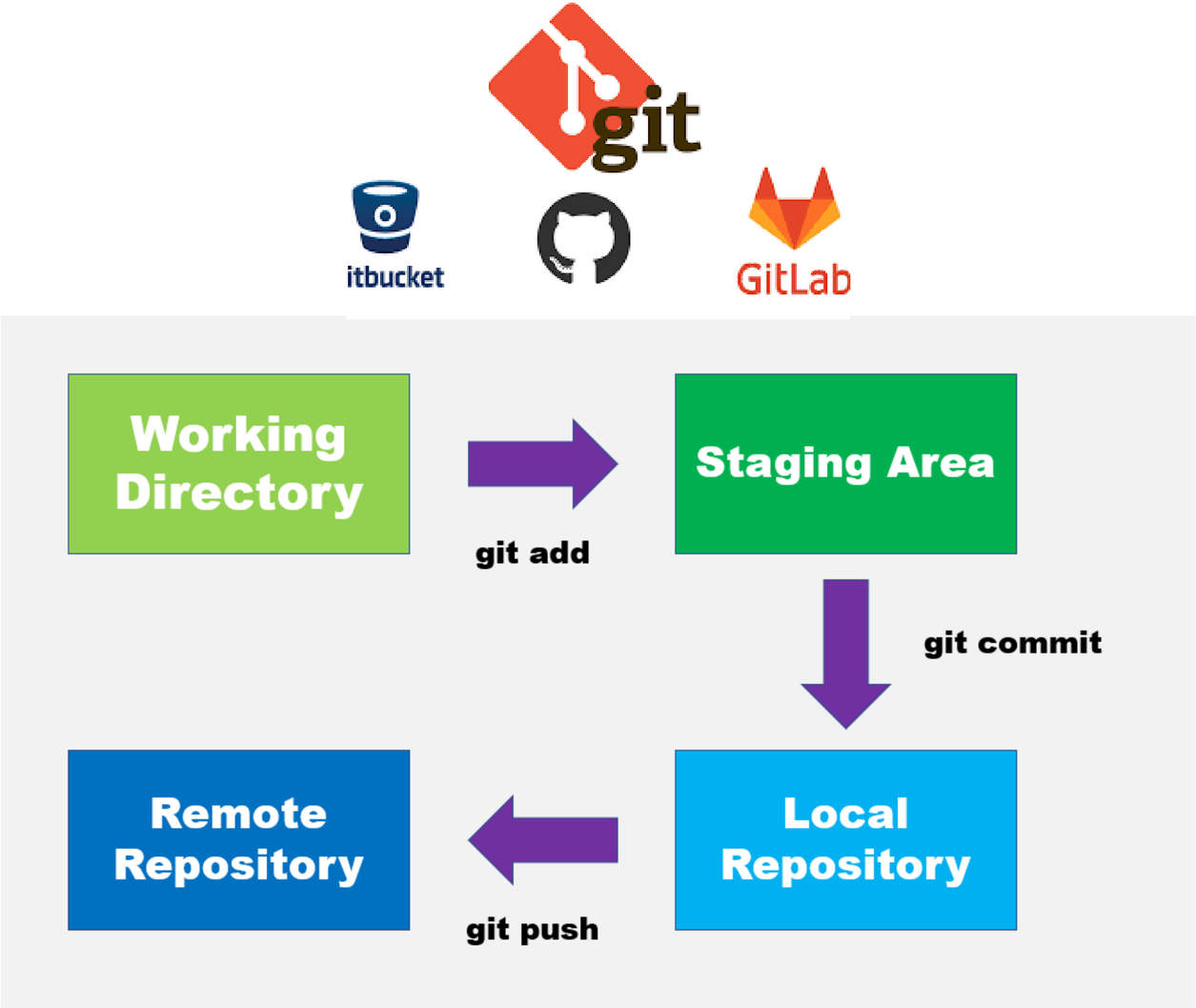
- A storage space for your project and its history
- Contains source code and metadata

Key Components:

- Working Directory
- Staging Area
- Local Repository

Remote Repositories:

- GitHub
- GitLab
- Bitbucket



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GIT SETUP

Initial Configuration:

- Set username with proper role (Owner, Developer, Maintainer)
- Set email
- Configure default branch

Best Practices:

- Use meaningful commit messages
- Keep commits small and focused
- Use .gitignore for:
    - Large datasets
    - Temporary logs
    - Environment files

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BRANCH STRATEGY

Purpose:

Define a clear workflow to manage development, testing, and releases safely and efficiently.

Main Branches:

main

- Production-ready code
- Stable and protected
- Only merged via Pull Request

develop

- Integration branch
- Contains latest completed features
- Used for testing before production

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BRANCH STRATEGY

Supporting Branches

feature/<feature-name>

- Used to develop new functionality
- Created from develop
- Merged back into develop

release/<version>

- Prepare production release
- Bug fixes and final validation
- Merged into main and develop

hotfix/<issue-name>

- Critical production fixes
- Created from main
- Merged into both main and develop

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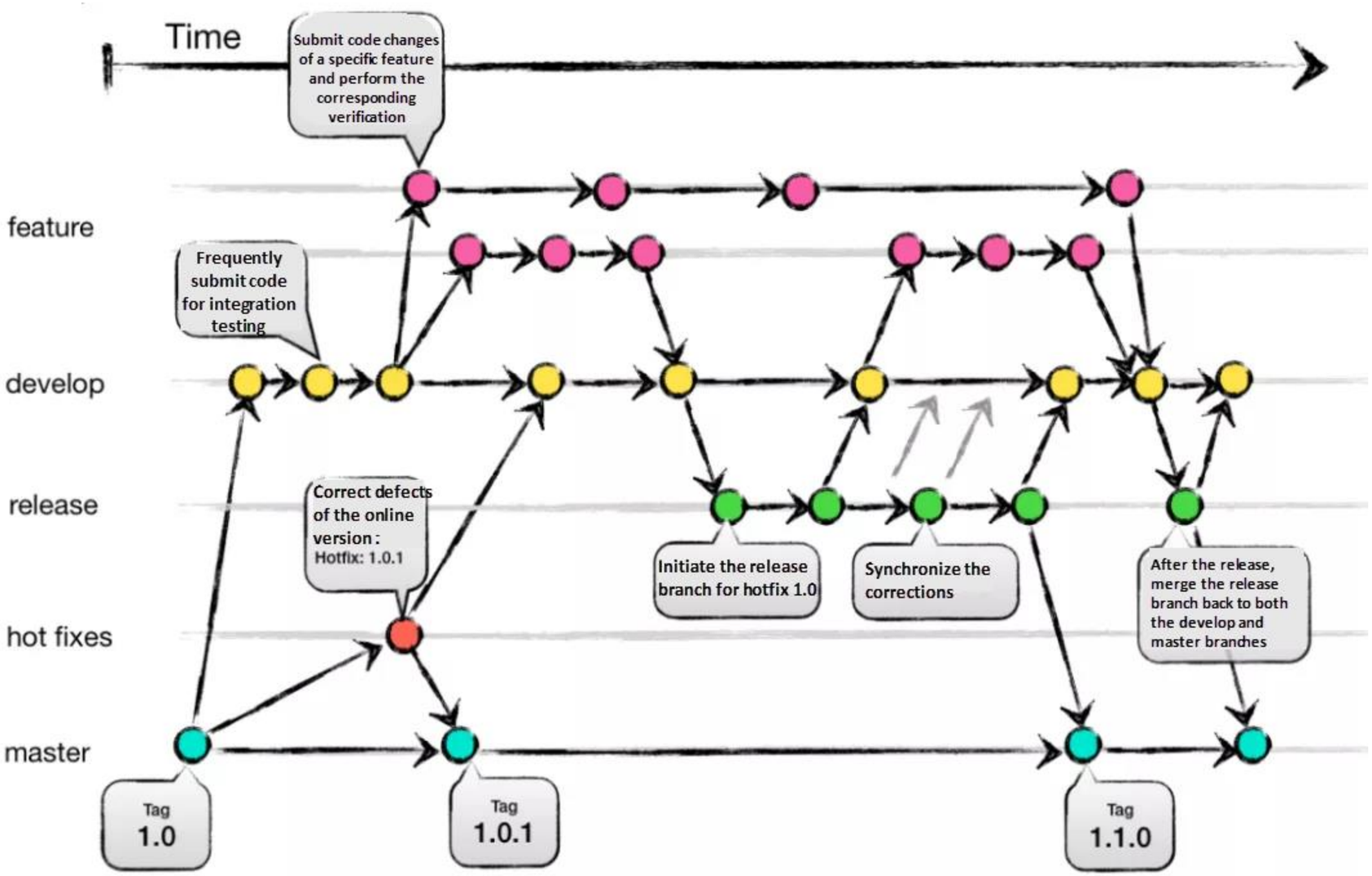
BRANCH STRATEGY



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APPLYING AI ACROSS THE GIT WORKFLOW

Version Control

- AI-generated commit messages
- Automatic change summarization
- Intelligent documentation generation

Managing Repositories

- AI-powered pull request reviews
- Automated bug detection in PRs
- AI-generated unit and integration tests
- Code quality and duplication analysis

GIT Setup & CI/CD

- Intelligent test prioritization
- Risk analysis for branches before merge
- Smart pipeline execution optimization

Branch Strategy

- Merge conflict prediction
- AI-assisted conflict resolution suggestions
- Automated release validation

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GIT WORKSHOP

WORKSHOP

Initial Steps:

1. Install Git
2. Configure Git locally
3. Create a local repository
4. Create and manage branches
5. Connect to a remote repository
6. Push changes to remote

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WORKSHOP

Scenario: "Simple Calculator API"

Build a very small calculator program that:

- Adds two numbers
- Subtracts two numbers
- Later introduces a bug
- Applies a hotfix
- Prepares a release version

The focus is NOT the code complexity, the focus is the **Git workflow**.

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SUMMARY

1. Install Git
2. Configure identity
3. Set up SSH
4. Clone or initialize repo
5. Follow standard workflow
6. Use branches for features/bugfix/hotfix

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